

# Midterm 1 Part b

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## I. QUESTION (16 POINTS)

Assume that you have a 2 dimensional 2 class dataset given below:

| $x_1$ | $x_2$ | C |
|-------|-------|---|
| 1     | 2     | 1 |
| 3     | 4     | 1 |
| 2     | 6     | 2 |
| 6     | 4     | 2 |

For the following algorithms, calculate estimated covariance matrix  $S$  (or matrices  $S_i$  depending on the algorithm).

- Quadratic Discriminant Analysis
- Linear Discriminant Analysis
- Naive Bayes
- K-means

## II. QUESTION (16 POINTS)

The following table shows the first, second midterm and final grades obtained for students in a database course.

| First ( $x_1$ ) | Second ( $x_2$ ) | Final ( $y$ ) |
|-----------------|------------------|---------------|
| 5               | 11               | 8             |
| 4               | 7                | 9             |
| 3               | 5                | 12            |

Derive a linear equation for the prediction of a student's third exam grade based on the student's first and second grades in the course. (Hint: Do a multivariate linear regression where  $x_1$  and  $x_2$  are the two midterm grades and  $y$  is the final grade. The function will be  $y = ax_1 + bx_2 + c$ )

## III. QUESTION (14 POINTS)

Given the following error rates for a four feature problem, what will be the result of the following attribute subset selection procedures?

- Stepwise forward selection
- Stepwise backward elimination

Error Rates: ( $F_1$ ), 0.23; ( $F_2$ ), 0.25; ( $F_3$ ), 0.36; ( $F_4$ ), 0.24; ( $F_1, F_2$ ), 0.18; ( $F_1, F_3$ ), 0.19; ( $F_1, F_4$ ), 0.20; ( $F_2, F_3$ ), 0.16; ( $F_2, F_4$ ), 0.29; ( $F_3, F_4$ ), 0.22; ( $F_1, F_2, F_3$ ), 0.39; ( $F_1, F_2, F_4$ ), 0.45; ( $F_1, F_3, F_4$ ), 0.24; ( $F_2, F_3, F_4$ ), 0.36; ( $F_1, F_2, F_3, F_4$ ), 0.59.